ROLE CARD

Sensor

Build step 1

Your part: BME688 sensor board (the small board with two STEMMA QT plugs)

It measures the air around it and reports temperature and humidity when the microcontroller asks.

REFERENCE GUIDE

- It measures four things: temperature, humidity, air pressure and gas. We use temperature and humidity only.
- It talks over I2C. That is two signal wires plus power and ground, all inside one plug-in cable.
- It has to touch open air. Sealed inside the box, it measures the box, not the room.

ONLY YOU KNOW. THE TEAM NEEDS THIS.

- Its I2C address is **0x77**. A few boards use 0x76. The code tries both.
- Humidity reacts to a breath in seconds. Temperature is slow. A warm hand works better than breath for temperature.
- It reads a little warm because it sits near the circuit board. The code has a setting for this: `TEMP_OFFSET_C`.
- Its two STEMMA QT ports are the same. One goes to the microcontroller. The other passes the bus on to the OLED screen.

YOU GIVE

Temperature (°C) and humidity (%) to Microcontroller + Code.

YOU NEED

Power and the I2C bus from Microcontroller + Code, through one cable. A spot at the vent from Mechanical. Tell Human Interaction which port is free.

YOUR BUILD TASK · STEP 1

With Microcontroller + Code: plug one end of a STEMMA QT cable into the sensor and the other end into the microcontroller's STEMMA QT port. The plug fits one way only. Do not force it.

IF SOMETHING GOES WRONG

The console says `Sensor not found - check the cable and the address`: reseat both ends of the cable.

CHECK YOURSELF

Why might our device read 1 to 2 °C warmer than the thermostat on the wall?

Outcomes: (1) Communicate effectively with engineers on other teams. (2) Collaborate across disciplines to build and improve a prototype. (3) Troubleshoot and document a prototype design.

ROLE CARD

Microcontroller + Code

Build step 1, 2 and 5

Your part: Adafruit QT Py RP2040, and `code.py` on the CIRCUITPY drive

The brain and its rules. It reads the sensor, decides LOW / OK / HIGH, and drives the screen and the color light, twice a second.

REFERENCE GUIDE

- The board runs CircuitPython. When it gets power it runs `code.py` on its own.
- Each reading has a good range between a minimum and a maximum: the set points. Below is LOW, above is HIGH, in between is OK.
- One USB-C port programs it, powers it and charges the battery.

ONLY YOU KNOW. THE TEAM NEEDS THIS.

- The STEMMA QT connector is the I2C bus we use. The edge pads are a separate bus. Nothing goes there.
- Power comes in on the **5V** pin. The **3V** pin is an output only. Never connect power to it. Leave the back jumper open.
- Set points now: `TEMP_MIN_C = 20.0`, `TEMP_MAX_C = 26.0`, `HUM_MIN_PCT = 40.0`, `HUM_MAX_PCT = 60.0`.
- Light rule: GREEN both OK. BLUE something LOW, nothing HIGH. RED anything HIGH. HIGH wins.
- At startup the console prints the I2C scan. **0x3c** and **0x77** means the wiring is right.

YOU GIVE

Power and the I2C bus to Sensor and Human Interaction. The LOW / OK / HIGH decision and the color.

YOU NEED

The 5V line at the 5V block and ground at the GND block, from Power. Readings from Sensor. What to show and what data to keep, from Human Interaction.

YOUR BUILD TASK · STEP 1, 2 AND 5

Step 1: take the sensor cable into the STEMMA QT port. Step 2: red wire from your 5V pin into the 5V block, black from your GND pin into the GND block. Step 5: plug in USB-C. Set the four set points for the application your team chose, and save. Then breathe on the sensor and watch the light.

IF SOMETHING GOES WRONG

The light never changes: check that the file saved and the console shows new readings.

CHECK YOURSELF

Humidity is LOW and temperature is HIGH at the same time. What color is the light, and why?

Outcomes: (1) Communicate effectively with engineers on other teams. (2) Collaborate across disciplines to build and improve a prototype. (3) Troubleshoot and document a prototype design.

ROLE CARD

Power

Build step 2

Your part: LiPo battery, MicroLipo charger, power board (four screw terminals and a diode), slide switch

It gets energy to the microcontroller from USB or from the battery, and charges the battery when USB is plugged in.

REFERENCE GUIDE

- One USB-C port does everything: the QT Py's port. The charger's own USB port is not used.
- Battery path: battery → charger → BAT block → diode → switch → the 5V line.
- The diode is a one-way valve. With USB plugged in, it stops the battery from pushing back onto the line.

ONLY YOU KNOW. THE TEAM NEEDS THIS.

- The power board has four blocks: **SW** and **BAT** on top, **5V** and **GND** on the bottom. In BAT, 5V and GND both screws are tied together, so either screw works. The two SW screws are separate.
- Wire colors: **yellow** is the battery (charger BAT pad to the BAT block). **Red** is 5V (charger 5V pad to the 5V block). **Black** is ground (charger GND pad to the GND block).
- Yellow never goes in the 5V block. That skips the diode and ties the battery straight to USB 5V. Red in the GND block shorts 5V to ground.
- The switch's two wires go in the SW block, either way round. The diode's stripe points up, toward the SW block.
- 5V line: about 4.6 to 4.7 V on USB, 3.4 to 3.9 V on battery. Battery: 3.7 V LiPo, 2000 mAh.

YOU GIVE

The 5V line at the 5V block and ground at the GND block, ready for Microcontroller + Code.

YOU NEED

Leave one free screw in the 5V block and one in the GND block for the QT Py's red and black wires.

YOUR BUILD TASK · STEP 2

Switch OFF. Battery into the charger's JST port. Then from the charger: yellow BAT into the BAT block, red 5V into the 5V block, black GND into the GND block. Switch wires into the SW block. Tighten and tug each wire.

IF SOMETHING GOES WRONG

Device dead on battery but fine on USB: check the switch, then that yellow is in the BAT block.

CHECK YOURSELF

Why use a diode here instead of a plain wire from the battery to the 5V line?

Power: wiring

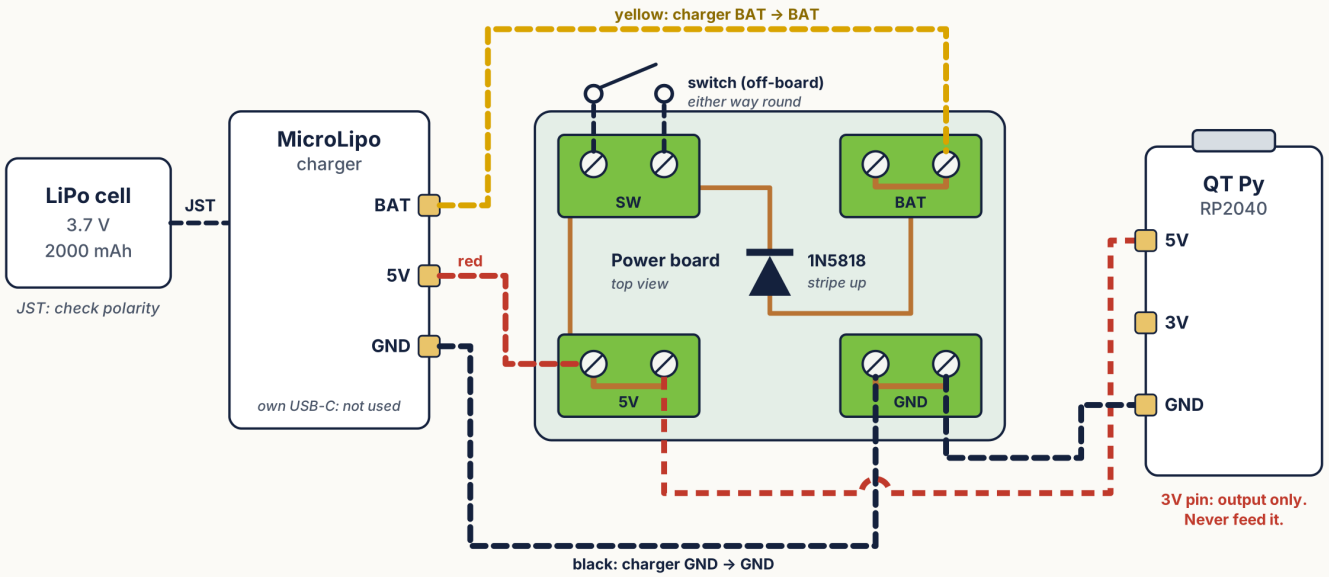
Build step 2

The dashed wires are your step. Check each against the labels before you tighten.

Power chain

One USB-C port (the QT Py's) programs the board, powers it and charges the battery.

Dashed wires are participant connections. Copper bar: both screws in BAT, 5V and GND are tied. SW screws are separate.



USB in: USB → QT Py 5V → 5V block → charger. Diode blocks the battery.

Battery: BAT → diode → SW → switch → SW → 5V block → QT Py 5V.

DESIGN QUESTION

Charge at 100 mA or 500 mA (the charger's rate jumper)? What does each cost in charge time, and what does the battery datasheet allow?

Outcomes: (1) Communicate effectively with engineers on other teams. (2) Collaborate across disciplines to build and improve a prototype. (3) Troubleshoot and document a prototype design.

ROLE CARD

Human Interaction

Build step 3

Your part: The screen, the color light, light pipes, and how people use the device

It decides how people use the device and what they need from it.

REFERENCE GUIDE

- Start with the question: what environment are we trying to measure, and who needs to know about it?
- The answer drives what people see and touch: the screen, the lights, the shape of the case, where it sits, what data it keeps.

ONLY YOU KNOW. THE TEAM NEEDS THIS.

- The OLED's I2C address is **0x3C**. It must differ from the sensor's 0x77, or the board cannot tell them apart.
- The screen plugs into the sensor's second STEMMA QT port, not into the microcontroller.
- Two light pipes: one over the QT Py's NeoPixel, one over the charger's charge LED. Each sits in a 5 × 5 mm square hole in the cover, right over its LED. A pipe that misses its LED stays dark.
- On battery there is no laptop console. The screen and the lights are the only readout.

DESIGN QUESTION

What environment are you trying to measure? Who uses this device, where, and what do they need to know? Decide what goes on the screen, what goes on a light, what data is worth keeping, and what shape the device should be.

YOU GIVE

The requirements: what people need to see, what data matters, where the device lives. Then the screen and lights themselves.

YOU NEED

The sensor's free port from Sensor. Microcontroller + Code to draw the screen and keep the data. Mechanical to build the shape, window and light pipe holes.

YOUR BUILD TASK · STEP 3

Plug a STEMMA QT cable from the sensor's free port into the OLED. Seat the OLED behind the display window. Fit the light pipes into their holes over the LEDs.

IF SOMETHING GOES WRONG

Screen blank: reseat the OLED cable. Light pipe dark: check it sits right over its LED.

CHECK YOURSELF

The device moves to a greenhouse. What would you change on the screen, and why?

Outcomes: (1) Communicate effectively with engineers on other teams. (2) Collaborate across disciplines to build and improve a prototype. (3) Troubleshoot and document a prototype design.

ROLE CARD

Mechanical

Build step 4 and 6

Your part: The 3D-printed case

It protects the parts and puts each one where it can do its job.

REFERENCE GUIDE

- Every opening has a job: a vent for the sensor, a window for the screen, holes for the light pipes, an opening for USB-C, a bay for the battery.
- The sensor mounts under the vent slots in the cover, away from the warm board.
- Cables need strain relief, so a tug does not pull a wire out of a terminal.

ONLY YOU KNOW. THE TEAM NEEDS THIS.

- In the cover: the sensor vent (five slots), the display window, and two 5 × 5 mm light pipe holes. In the body: the battery bay (the walled-off section), the USB-C opening (8 × 14 mm, top left) and the switch hole (10 mm round, top right).
- The cover closes with four stainless steel M3 socket head cap screws at the corners. The boards mount with nylon M2.5 screws.
- The lid goes on last, after the test passes. Opening it again costs time.
- The battery sits in its bay with its wire leaving toward the charger. Nothing pinched.

YOU GIVE

A fixed place for every part.

YOU NEED

Sensor at the vent (Sensor). The shape, the window and the light pipe holes from Human Interaction. USB-C port at its opening (Microcontroller + Code). Battery in the bay (Power).

YOUR BUILD TASK · STEP 4 AND 6

Step 4: seat each part in the case, route the cables, check nothing is pinched. Leave the lid off. Step 6: close the case after the test passes.

IF SOMETHING GOES WRONG

Temperature drifts up after the lid goes on: the sensor is too close to the board or the vent is blocked.

CHECK YOURSELF

If the sensor sat next to the microcontroller with no vent, which reading would be wrong, and in which direction?

Outcomes: (1) Communicate effectively with engineers on other teams. (2) Collaborate across disciplines to build and improve a prototype. (3) Troubleshoot and document a prototype design.